Fig. 4: Extracting features from the spectrogram

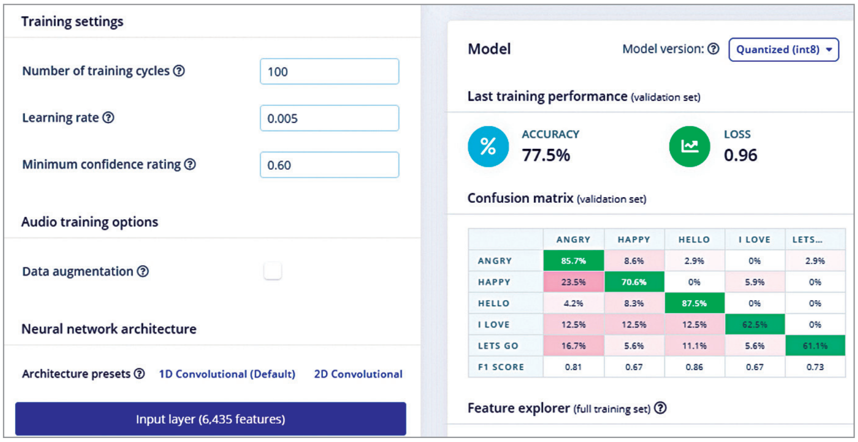

Fig. 5: Accuracy output during ML model testing

them based on different emotions (angry, glad, hungry) or expressions ("Let's go", "I love you").

Note. Before proceeding with sound data capture, put the AIY Voice bonnet onto the Raspberry Pi as shown in Fig. 1.

ML model

Select the learning and processing blocks for training the ML model. Here we use Spectrogram for the processing block and Keras for the learning block. Using these, extract different audio parameters for training the ML model to learn from datasets. Then test the model and keep refining it until you are satisfied with its accuracy.

To deploy the ML model, go to the deploy option and select Linux board. Install it and clone the SDK of edge impulse.

```
sudo apt-get install libatlas-base-dev
libportaudio0 libportaudio2 libportaudi
ocpp0 portaudio19-dev
pip3 install edge_impulse_linux -i https://
pypi.python.org/simple
pip3 install edge_impulse_linux
git clone https://github.com/edgeimpulse/
linux-sdk-python
```

Coding

Create a .py file called animal_translate and import espeak in the Python

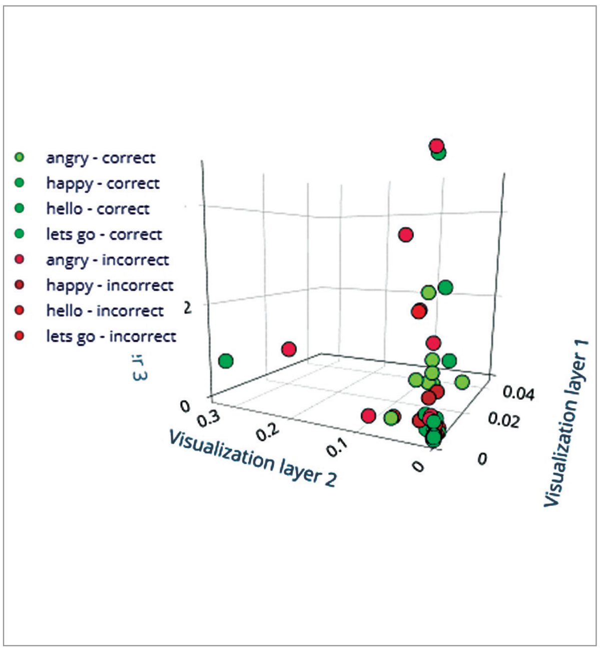

Fig. 6: 3D Graph showing correct/incorrect recognition during testing of ML model

```python
import os
import sys, getopt
import signal
import time
from espeak import espeak
from edge_impulse_linux.audio import AudioImpulseRunner

runner = None

def signal_handler(sig, frame):
    print('Interrupted')
    if (runner):
        runner.stop()
    sys.exit(0)

signal.signal(signal.SIGINT, signal_handler)

def help():
    print('python classify.py <path_to_model.eim> <audio_device_ID, optional>' )

def main(argv):
    try:
        opts, args = getopt.getopt(argv, "h", ["--help"])
    except getopt.GetoptError:
        help()
        sys.exit(2)

    for opt, arg in opts:
        if opt in ('-h', '--help'):
            help()
            sys.exit()

    if len(args) == 0:
        help()
        sys.exit(2)

    model = args[0]

    dir_path = os.path.dirname(os.path.realpath(__file__))
    modelfile = os.path.join(dir_path, model)
```

Fig. 7: Python code